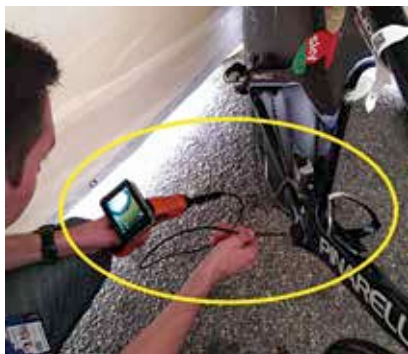




EPO isn't the only thing cyclists use to cheat their way to the podium

hidden away within carbon-fibre wheels to generate a boost of up to 60 watt.

OLYMPIC GRADE CHEATING

Nothing is sacred in the quest to win at any cost. Not even the Olympics. Even as the never-ending war between the performance-enhancing drug industry and World Anti-Doping Agency (WADA) rages on, geeks have managed to slip through the cracks and give athletes sophisticated equipment that endows them with an unfair advantage. Informally known as technology doping, this is a relatively new branch of high-tech cheating that leverages advancements in nanotechnology, exotic engineering materials, and aerodynamics to enhance athletic performance. It's ironic how on the one hand athletes are given severe penalties and even lifetime bans for doping, but on the other hand technology doping flies right under the radar.

The most egregious of these examples is the infamous LZR swimsuit introduced in the 2008 Olympics. Developed by Speedo, the specialised suit employed a combination of high-tech materials and cutting-edge computer-aided design to make swimmers slippery and greatly reduce water resistance. What's more, the suit's design not only allowed better oxygen flow to muscles, thus enhancing performance, but it also trapped air to aid buoyancy. To put its efficacy into perspective, swimmers wearing the swimsuit were responsible for 23 out of the 25 world records set during the 2008 Olympics. If you didn't have one, you were at a genuine disadvantage. The Olympic governing body FINA has since banned the outfit. American speed skaters took



The Speedo LZR swimsuit was so good that it was responsible for 43 world records in 2009

the same route by adopting a specialised uniform designed by fashion label Under Armour in conjunction with Lockheed Martin – you know, the same guys responsible for the SR-71 Blackbird, F-22 and F-35 fifth-generation jet fighters! Obviously the guys who make supersonic stealth fighter jets know a thing or two about aerodynamics. So far the sports authorities haven't objected to these suits, but it could just be a matter of time once they start breaking world records by the dozen. It's the same story with BMW pairing up with the US bobsled team to create an all-carbon fibre, lightweight and aerodynamically efficient sled that is bound to give it the decisive edge.

With geek cheats that make athletes go fast, we have one aspect of the Olympic motto figured out. But what about the one dealing with going higher? Well, the specialised shoe maker



You can't go wrong when Lockheed Martin designs your skating uniform

Athletic Propulsion Labs have you covered in that regard. Their \$300 flagship basketball shoe dubbed 'Concept 1' received the ultimate endorsement when the NBA banned it in 2010. The shoe uses a combination of specialised foam, air vents, and springs to create what the designers describe as the "springboard effect". The shoe is claimed to allow basketball players to jump several inches higher than usual. Moreover, the shoe's unique suspension system and energy transfer capability is touted to shave off up to nine seconds off the average runners mile time. No wonder NBA was quick to ban these high-tech kicks.



This \$300 shoe was banned by the NBA for making players faster and helping them jump higher

MOTORSPORTS: CHEATING OR INNOVATION?

Motorsport easily takes the cake when it comes to cheating. And that's partly because it's a sport where engineers are nearly as important, if not more important than the drivers themselves. That's where the problem of cheating transforms into a rather abstract concept. The job of an automotive engineer, in this case, is to push the envelope of technology and employ every bit of lateral thinking to reinterpret the rules to work around the performance limiting regulations. The very concept of cheating in this respect is just another way of looking at the pursuit of innovation and engineering ingenuity. And there's a lot of ingenuity going around in this sport.

Very few automotive engineering endeavours have affected motorsport as profoundly as aerodynamics. To put this into perspective, more than half the fuel consumed by a regular road car travelling at a speed of just 60 kmph is spent to overcome the air resistance or drag.